

[STRATEGIC INFRASTRUCTURE REVIEW]

MAHA STRATEGIES

AI Infrastructure Report

Architecture, readiness, improvement options, and paths to revenue

REPOSITORY SNAPSHOT 32976b3 | 08 AUGUST 2026

PUBLIC SURFACE CHECK 08 AUGUST 2026 | ASIA/COLOMBO

Prepared for strategic brainstorming and commercial prioritization

MAHA STRATEGIES LLC

SCOPE AND EVIDENCE STANDARD

How to read this report

Purpose

Create one decision document that distinguishes what is implemented, what is publicly reachable, what is commercially enabled, and what still requires customer evidence. The report is designed to support a focused improvement and revenue workshop.

Evidence classes

Class	Meaning	Examples in this report
Code-verified	Observed in the current repository at commit 32976b3.	Routes, controls, schemas, migrations, SDK methods, workflows.
Benchmark-verified	Reproducible artifact or documented hardware run with stated limits.	MCRB-1, A10G tensor-network and SE(3) baselines.
Live public check	Unauthenticated HTTP or read-only conformance probe performed on 2026-08-08.	Public pages, discovery files, protected 401s, x402/Bazaar drift.
Operator-reported	Evidence supplied during the working session but not re-executed here.	Successful Base Mainnet settlement and wallet receipt.
Recommendation	A proposed action or economic scenario, not a statement of current fact.	Packaging, pilot pricing, sequencing, and revenue models.

No production secrets were read. Protected readiness endpoints were tested only for the expected unauthenticated 401 behavior. No paid x402 call, Stripe charge, database mutation, or deployment was performed for this report.

Current repository snapshot

OPENAPI PATHS 35 Documented public API paths	DB MIGRATIONS 66 Append-only Supabase migration files	CI WORKFLOWS 11 Quality, E2E, health, migration, and recovery
TEST FILES 78 Node test modules	VALIDATED TESTS 420/420 Last production-build validation in this session	WORKTREE Clean main equals origin/main at report start

STRATEGIC CONCLUSION

Executive summary

Maha is no longer blocked by core engineering readiness.

The platform contains a credible set of production-grade primitives: a benchmarked context compiler, a tenant-aware MCP gateway, bounded GPU jobs, provenance/audit products, dual payment rails, public SDKs, and unusually mature release controls for a small operator. The binding constraint is now demand concentration and proof of paid use, not missing infrastructure.

The highest-probability revenue wedge is the Context Compiler because it has the cleanest unit transaction, the strongest public benchmark, a zero-install playground, a \$0.001 x402 path, and a conventional API-key path. The strongest higher-value enterprise wedge is the MCP Gateway because governance, allowlists, audit evidence, and tenant isolation map to budget owners - but it still needs a narrowly scoped paid design partner and a concise operational proof package.

The GPU APIs add technical depth and may win specialist workloads, but they are not the next general revenue engine. The tensor-network and SE(3) engines have honest A10G baselines; the general QUBO route is exposed while its checked-in promotion document still says private and its earlier baseline failed. That contradiction should be resolved before active selling.

Immediate priorities

Rank	Action	Why now	Done when
P0	Refresh Bazaar metadata and make x402 drift a release check	The live v2 endpoint passes, but Bazaar currently serves a different declaration digest.	Doctor reports no stale-metadata warning after a fresh bounded settlement/index refresh.
P0	Choose one commercial wedge for 30 days	Broad capability creates buyer confusion and fragmented proof.	One buyer profile, one workflow, one offer, one success rubric, one landing path.
P1	Run 3-5 customer-context pilots	MCRB-1 proves benchmark behavior, not customer outcome or willingness to pay.	At least one repeated workflow shows net cost or quality benefit and converts to paid use.
P1	Canonicalize MCP gateway ownership	Two gateway/registry paths increase security-review and support burden.	One documented public contract, one tenant registry, one audit schema, explicit compatibility plan.
P1	Close GPU evidence inconsistencies	Public QUBO exposure conflicts with its promotion runbook.	Every public engine has current hardware evidence, E2E results, limits, and matching documentation.
P2	Instrument acquisition-to-paid conversion	Traffic and listings do not reveal which path creates revenue.	Weekly funnel by discovery source, first successful call, repeated usage, and paid retention.

ANALYST ASSESSMENT - NOT A CUSTOMER SLA

Infrastructure scorecard

Dimension	Score	Strength	Primary gap
Application architecture	8/10	Clear serverless boundaries, typed contracts, fail-closed routes.	Consolidate duplicated/legacy surfaces.
Security controls	8/10	Tenant binding, SSRF checks, encrypted secrets, RLS, replay guards.	Independent threat model and external test.
Payments and ledgers	8/10	Stripe idempotency, append-only credits, x402 v2, chain confirmation.	Unified reconciliation and live revenue dashboards.
Reliability operations	9/10	Release health, LKG artifacts, migration evidence, rollback drills.	Prove recent controlled alert and restore rehearsal evidence.
Developer experience	8/10	OpenAPI, TS/Python SDKs, CrewAI/LangChain, recipes, playground.	One canonical quickstart and integration telemetry.
Benchmark credibility	8/10	MCRB-1 raw records and honest GPU baselines.	Customer workload results; QUBO evidence cleanup.
Machine discovery	7/10	Agent card, offers, llms.txt, Bazaar, package registries.	Resolve current Bazaar drift; monitor registry freshness.
Commercial focus	5/10	Multiple working monetization paths.	Concentrate message, validate willingness to pay.
Customer proof	3/10	Pilot design exists; no verified external recurring use in repo.	Reference customer, case study, retention evidence.

Overall interpretation

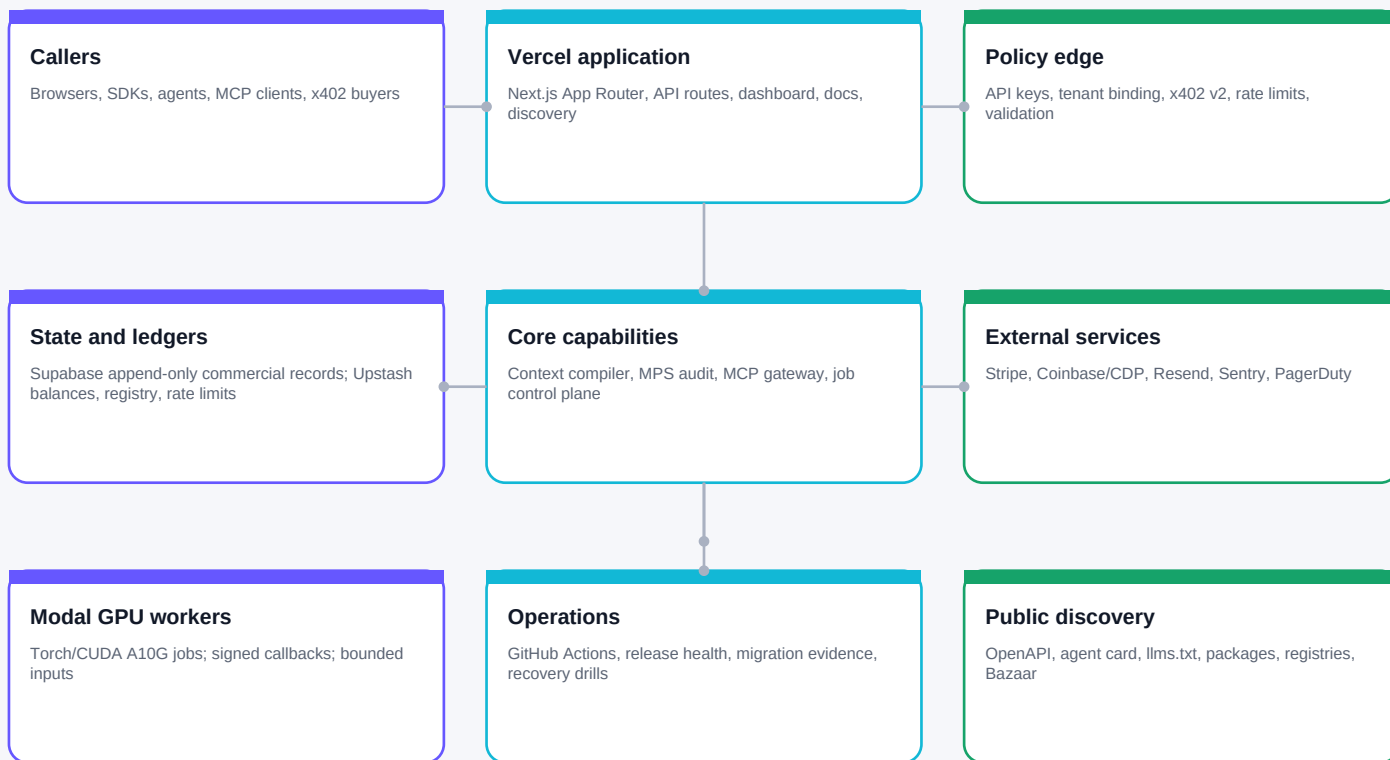
Engineering readiness is approximately 80-90% for controlled production use. Commercial readiness is closer to 50-60% because repeat usage, buyer urgency, and channel conversion remain unproven. Building more horizontal capability is unlikely to close that gap.

What is actually live or publicly reachable

Surface	Observed	Interpretation
Homepage, Developers, Context Compiler, MCP Gateway, MCRB-1	HTTP 200	Core public narrative and evidence pages are reachable.
OpenAPI, agent card, offers, MCP contract, llms.txt, sitemap	HTTP 200	Human and machine discovery surfaces are reachable.
Billing, revenue, observability readiness without token	HTTP 401	Protected operator diagnostics correctly reject anonymous callers.
Context Compiler unpaid probe	HTTP 402, x402 v2	Live challenge and crawler request are valid.
Coinbase Bazaar record	Found, but digest differs	Listing exists; metadata refresh is incomplete.
/.well-known/x402	HTTP 404	Not a protocol failure; optional standalone manifest is absent.

CURRENT PRODUCTION TOPOLOGY

System architecture



The architecture is intentionally split by responsibility. Vercel hosts the public site and request control plane. Supabase holds durable append-only commercial and operational records. Upstash supplies low-latency balances, rate limits, and parts of the API/MCP registry. Modal executes bounded Torch/CUDA workloads. Stripe and x402 provide separate human-authorized and autonomous payment paths.

Architectural strength

The platform does not trust a payment, callback, API key, upstream MCP URL, or retry merely because it is syntactically valid. Most state changes are bound to an idempotency key, tenant identity, hash, signed event, or on-chain confirmation.

Architectural tension to resolve

- Supabase and Upstash are both legitimate dependencies, but ownership boundaries are difficult to explain externally. Publish a one-page data-flow and system-of-record table.
- The MCP capability has multiple route families and registry implementations. Compatibility may be intentional, but security reviewers will ask which is canonical.
- The OpenAPI document is titled around MPS even though it now describes a broad developer platform. Rename and version the product-level contract without breaking paths.
- The application is one large deployment. That is efficient today, but GPU jobs, payment webhooks, public content, and operator APIs share one release blast radius.

WHAT EXISTS AND HOW MATURE IT IS

Capability portfolio

Capability	Status	Access	Evidence	Commercial role
Context Compiler	Production	API key + x402	MCRB-1; live recipe	Best immediate usage wedge
Context Pack Evaluator	Production API	API key	Contract/tests	Pilot acceptance layer
Enterprise MCP Gateway	Production v1	API key/tenant	E2E, controls, audit	Best enterprise wedge
MPS audit/preflight	Production	Free, prepaid, Stripe	Audit engine/tests	Document assurance niche
Tensor-network QUBO	Production API	Credits	A10G p95 80.840 ms	Specialist technical offer
SE(3) registration	Production API	Credits	A10G p95 108.604 ms	Specialist technical offer
Parallel QUBO/Ising	Public API	Credits	Earlier candidate failed 500 ms gate	Evidence/doc gap
Audit export	Production	API key	CSV/PDF E2E	Enterprise trust feature
Developer billing	Production code	Stripe packs/subscriptions	Webhook/idempotency tests	Needs conversion proof
SDKs/integrations	Published	npm/PyPI	TS 0.2.0; Python 0.1.0	Acquisition enabler
x402 Doctor	Operational tool	Open-source candidate	CLI, SARIF, tests	Ecosystem credibility

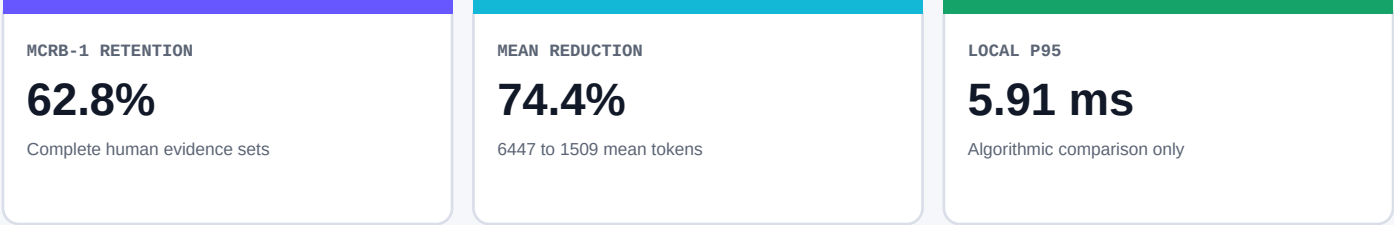
Portfolio rule

Do not market every capability equally. Lead with one measurable job, use the MCP Gateway as the enterprise expansion path, and treat GPU APIs plus MPS as targeted offers triggered by a buyer's concrete workload.

STRONGEST NEAR-TERM PRODUCT

Context Compiler

The Context Compiler accepts 1-8 source documents, an explicit task, and a fixed output budget. It performs deterministic extractive selection with BM25 or keyword scoring, removes exact duplicate passages, returns source and passage identities, publishes hashes, and exposes explicit retention warnings. Standard and enterprise payload ceilings are 525 KB and 1.2 MB respectively.



Why it is commercially attractive

- The buyer can understand the transaction: pay \$0.001 to reduce downstream input spend and preserve passage-level provenance.
- The product can be tried without installation, called with conventional API keys, or purchased autonomously through x402.
- The benchmark is reproducible and publishes the cohort, per-case records, aggregate results, method boundaries, and limitations.
- It is model-neutral and sits before the expensive model, making it complementary to OpenAI, Anthropic, CrewAI, LangChain, and custom RAG stacks.

Where the product still needs proof

- MCRB-1 measures evidence retention, not generated-answer correctness, user value, or production API latency.
- QASPER research papers are not representative of every enterprise corpus. Customer-specific labeled traces are required.
- The \$0.001 fee is compelling per call but requires extremely high call volume to become material revenue on its own.
- A compiled pack can omit necessary evidence. The product should make acceptance thresholds and failure handling first-class.

Recommended positioning

Do not sell generic 'compression.' Sell a bounded pre-inference step for large RAG/tool payloads where the customer can label required evidence and measure cost per successful task.

MEASURED ADVANTAGE AND REMAINING HEADROOM

MCRB-1 benchmark interpretation

Complete evidence-set retention at a 2,048-token budget



Question	Measured answer	Commercial implication
Does ranking beat naive positional selection?	Yes. BM25 retained 62.8% complete sets vs 25.6% front and 20.4% tail.	A real differentiator for evidence distributed throughout long documents.
Does compilation preserve every required fact?	No. 37.2 percentage points remain below the oracle ceiling.	Acceptance gates and fallback to raw context are essential.
Are citations traceable?	100% for every extractive method in this protocol.	Traceability is a design property, not evidence completeness.
Is the savings economic?	\$0.014815 gross input cost avoided at a \$3/M-token reference rate; \$0.013815 net after fee.	The fee captures about 6.75% of gross avoided input cost for the benchmark mean.
Does this beat LLM summarization?	Not established. Generative summarizers were excluded to avoid model-dependent judging.	Publish a separate answer-quality benchmark before making that claim.

Next benchmark

Run a customer-oriented benchmark with three layers: exact required-fact retention, downstream answer/task quality, and total cost per successful result. Include at least one generative summarizer under a declared model and judge protocol, but keep that benchmark separate from MCRB-1.

PRODUCTION CAPABILITY WITH A DISCOVERY DEFECT

x402 payment rail

The production seller contract uses x402 v2 on Base Mainnet, exact payment, USDC, and a 1,000-base-unit price equal to 0.001 USDC. The request path verifies the published requirement, guards against replay in Supabase, settles through the facilitator, optionally confirms the transfer on chain, and only then applies resource concurrency admission.

Control	Current behavior
Protocol	x402 v2; CAIP-2 eip155:8453; exact scheme.
Price	1,000 USDC base units = 0.001 USDC per Context Compiler call.
Replay	Payment identity is atomically claimed before settlement reuse is possible.
Chain check	Asset, payer, payee, minimum amount, receipt, and chain are independently corroborated when RPC is configured.
Capacity	A bounded Redis slot is acquired after payment; paid-at-capacity calls receive an explicit retry condition.
Discovery	Bazaar extension carries a complete input/output schema and realistic example.
Buyer safety	Executable recipe enforces a \$0.005 ceiling and expects exactly \$0.001 before signing.

Live finding: Bazaar metadata drift

On 2026-08-08, x402 Doctor found the production endpoint valid (HTTP 402, v2, crawler HTTP 402) and the Bazaar listing present. However, the live declaration digest f704...b7c5 did not match Bazaar digest e077...a85d. The listing can therefore describe an older contract until refreshed.

Recommended fix and guardrail

- Perform one bounded post-change settlement only when the discovery contract changes, then confirm the Bazaar digest matches the live digest.
- Run x402 Doctor in CI with fail-on-warning for the production canary after metadata-affecting releases.
- Add a release evidence record containing live digest, Bazaar digest, listing timestamp, transaction hash, and crawler status - never the wallet key.
- Propose the declaration-digest/version extension upstream so registries can detect stale seller metadata without relying on an unrelated paid call.

BEST HIGHER-VALUE ENTERPRISE WEDGE

Enterprise MCP Gateway

The gateway registers approved public HTTPS MCP upstreams under a tenant, encrypts supported upstream credentials, discovers tools through a bounded tools/list exchange, applies explicit method and tool allowlists, and records metadata-only outcomes. Tenant-wide settings bound requests per minute, timeout, failure threshold, and circuit cooldown.

Control layer	Implemented behavior	Commercial value
Tenant registry	Server records resolve under the caller's tenant; credentials are not returned.	Central inventory and revocation.
Tool discovery	Validated tools/list inventory; discovery does not authorize tools.	Operator sees actual exposed capabilities.
Allowlists	tools/call requires an exact approved name; methods are explicitly bounded.	Least privilege for agent actions.
SSRF containment	HTTPS-only registration, DNS checks, redirect limits, bounded sizes.	Reduces arbitrary outbound access risk.
SLA controls	Per-tenant rate, timeout, failure threshold, cooldown, circuit breaker.	Limits blast radius from unhealthy upstreams.
Audit	Server, method, tool, outcome, status, hash, and time; no arguments or bodies.	Investigable access without copying customer payloads.

The likely first enterprise offer

A 30-day governed MCP pilot for one team, 2-5 upstream servers, a named allowlist owner, baseline direct connections, and a weekly access/failure review. Charge for implementation and evidence, not raw proxy volume.

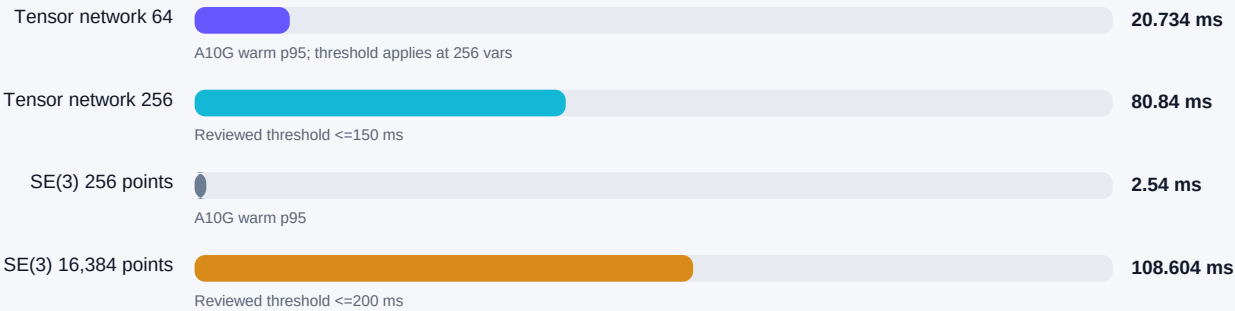
Gaps before enterprise scale

- Choose and document one canonical gateway path/registry. The codebase still contains multiple MCP route families and compatibility surfaces.
- Add upstream OAuth/token-exchange support only for a signed design partner; do not build speculatively.
- Private connectivity and controlled egress are necessary for high-assurance customers; application-level DNS checks are not the final network boundary.
- SSE/streaming remains unsupported. Measure real buyer demand before adding it, then design billing and audit semantics for long-lived sessions.
- Produce a shareable threat model, data-flow diagram, subprocessors list, retention statement, and incident-response summary.

TECHNICALLY CREDIBLE, COMMERCIALY SPECIALIZED

GPU optimization platform

Published warm solver p95 latency



Engine	Honest boundary	Evidence state	Revenue view
Tensor-network QUBO/Ising	Sparse models, <=256 variables, bounded bond dimension; heuristic above exact threshold.	7 warm A10G runs per case; objectives recomputed.	Offer to optimization teams with fitting problem sizes.
Weighted SE(3) registration	Known correspondences, rigid motion, 3-16,384 paired points.	Known transform recovered; residual and determinant verified.	Offer to robotics/vision prototypes needing a managed API.
Parallel QUBO/Ising	Heuristic; public route currently exists.	Checked-in A10 baseline failed 500 ms gate; promotion guide says private.	Do not actively sell until evidence and documentation converge.

Priority correction

The public QUBO/Ising route, OpenAPI, SDK, and the 'private/no endpoint' promotion document cannot all be current simultaneously. Resolve the contradiction and publish one authoritative status record before customer outreach.

Next engineering work only if demand exists

- Measure full enqueue-to-result p50/p95/p99, including Vercel, queue, Modal cold start, callback, and polling.
- Add cost per job and gross margin by problem size; current credit quotes are deterministic but not tied to published infrastructure cost.
- Create workload-specific examples with expected objective/residuals rather than generic random matrices.
- Keep x402 away from expensive asynchronous GPU paths until capacity, refunds, and post-payment admission are economically explicit.

TRUST LAYER WITH MULTIPLE ACQUISITION MODES

MPS, audit, and evidence products

MPS spans a free public audit, a paid preflight, prepaid headless audit credits, claim resolution, audit exports, and evidence-oriented educational content. The strongest design feature is not a truth claim; it is the explicit labeling of provenance status, boundaries, and recommended review action.

Surface	Buyer	Payment	Current role
Public /audit	Evaluator/editor	Free, rate-limited	Lead capture and product education.
MPS Preflight	Document owner	Stripe; public page states \$49	Human-authorized document review product.
Headless MPS audits	Agent or integration	Prepaid audit credits	Repeatable claim-level API workflow.
Audit export	Enterprise operator	Included with API relationship	CSV/PDF evidence and ledger trail.
Claim registry	Research reader/agent	Read-only	Evidence status and source discovery.

Commercial hypothesis

MPS is more likely to monetize through high-consequence workflows - reports, public claims, policy documents, investor research, regulated communication - than as a general-purpose fact checker. Buyers pay for a repeatable review record, not for an assertion that the model guarantees truth.

Improvements

- Publish one anonymized before/after case showing claim tagging, reviewer decisions, time saved, and failure boundaries.
- Unify free audit, paid preflight, and prepaid API language so buyers understand which output is advisory, operational, or human-reviewed.
- Measure free-to-paid conversion by source page and workflow, not total visits.
- Make export verification self-contained: hash algorithm, event ordering, schema version, and a verifier command.

STRONG PRIMITIVES; COMPLEX PRODUCT SURFACE

Identity, billing, and data systems

Subsystem	Implemented model	Key risk
API keys	Hash-only storage, one-time disclosure, rotate/revoke, tenant linkage.	User support when a key is lost; prevent secrets in browser persistence/logs.
Credits	Shared tenant pool with subscription and rollover buckets; append-only movements.	Credit economics are not intuitive across heterogeneous products.
Subscriptions	Builder \$20/10k monthly; Scale \$100/60k; monthly reset.	Value communication and usage fit are not yet proven.
Auto top-up	\$10 for 5k rollover credits below 1k threshold; explicit opt-in.	Off-session failures, customer notice, and retry policy.
Stripe	Signed webhooks, event idempotency, grant/reversal separation.	Many product-specific webhook endpoints and price maps.
x402	Per-call USDC settlement with replay guard and optional chain verification.	Registry freshness, wallet friction, and uncertain organic agent demand.
Supabase	Durable private ledgers and operational records; RLS/no public policies.	Migration evidence and backup/restore discipline must stay current.
Upstash	Balances, rate limits, environment namespaces, MCP/API registries.	Cross-store consistency and clear system-of-record ownership.

Design recommendation

Publish a customer-facing credit calculator. Credits are safe for internal metering, but buyers should see an estimated number of Context Packs, MCP calls, or bounded GPU jobs per plan - plus the exact assumptions.

Example capacity at current job quotes

At 256 variables, QUBO and SE(3) jobs quote 525 credits and tensor-network jobs quote 800 credits. A 10,000-credit Builder grant therefore represents about 19 QUBO/SE(3) jobs or 12 tensor-network jobs before other usage. A 60,000-credit Scale grant represents about 114 or 75. This arithmetic is code-derived; it is not a promise of workload fit or infrastructure cost.

WHAT IS DEFENSIBLE TODAY

Security and privacy posture

Area	Current control
Secret handling	Server-only environment variables; one-time credential disclosure; hash-only key storage.
Tenant isolation	Credentials resolve to tenant-scoped resources; MCP server records are tenant-bound.
Upstream safety	HTTPS-only URLs, no embedded credentials, DNS checks, redirect and size bounds.
Payments	Signed Stripe webhooks, x402 EIP-712 context, replay guard, optional on-chain corroboration.
Callbacks	Signed Modal worker callbacks; bounded job contracts and server-side credit reservation.
Data minimization	Context source text and packs are not stored by the compiler; gateway audit excludes payloads.
Telemetry	Sentry scrubbing, low-cardinality spans, bounded signed operational alerts.
Database	RLS with no public policies; append-only migrations and ledger conventions.

Highest-value security improvements

- Commission an external review focused on tenant isolation, webhook/replay correctness, MCP SSRF, and x402 payment-state ambiguity.
- Create a concise public security page covering data flows, retention, encryption, subprocessors, vulnerability reporting, and incident notification.
- Add controlled egress/private connectivity options before claiming suitability for sensitive enterprise MCP upstreams.
- Document key rotation ownership and recovery for Vercel, Supabase, Upstash, Modal, Stripe, Coinbase/CDP, Sentry, Resend, and PagerDuty.
- Define deletion and export semantics for tenant configuration, audit metadata, and commercial records while preserving legally required append-only evidence.

Legal boundary

x402 production capability does not answer tax, accounting, virtual-asset, sanctions, consumer-protection, or licensing questions. Preserve counsel's written advice and translate it into an operational checklist before expanding networks, assets, custody, or marketing claims.

A MAJOR PLATFORM STRENGTH

Day-2 operations and release safety

Control	Cadence / gate	Evidence
Quality	PR and push	Migration integrity, typecheck, lint, tests, x402 dry-run, production build.
Preview E2E	Deployment/manual	MCP register, proxy call, CSV export, PDF export.
Preview capacity	Daily + manual	Public/control-plane trend; bounded MCP spend only with confirmation.
Production release health	Every 15 minutes	Homepage, OpenAPI, billing readiness, observability readiness; LKG artifact.
Recovery drill	Weekly	Immutable LKG deployment remains ready and passes four checks.
Rollback	Reviewer-protected manual	Exact confirmation, deployment identity, E2E verification, 90-day evidence.
Rollback/restore rehearsal	Reviewer-approved maintenance	Both alias transitions and post-state verification.
Production migrations	Reviewer-protected per change	Dry run, drift, apply confirmation, post-apply convergence and health.
Environment isolation	Daily	Preview/Production credential assignment policy.

Assessment

The release and recovery system is more mature than the current customer base requires. Preserve it, but avoid adding new operational machinery unless it protects a real paid path or closes a demonstrated incident mode.

Remaining reliability work

- Maintain a dated index for every production migration and supervised rollback/restore rehearsal; do not rely only on expiring GitHub artifacts.
- Confirm PagerDuty escalation has a second recipient and periodically run the controlled failure/recovery test.
- Track full customer-path availability separately from public page availability: authentication, payment, ledger, worker, and callback.
- Define incident severity and response expectations that one operator can realistically sustain before promising enterprise SLAs.

BROAD COVERAGE, FRESHNESS RISK

Developer experience and discoverability

Surface	Status	Role
OpenAPI 3.1	35 paths, live JSON and interactive docs	Canonical HTTP contract.
TypeScript SDK	@mahastrategies/sdk 0.2.0	Node, Bun, Deno, browser, Edge.
Python SDK	maha-sdk 0.1.0	Base client plus LangChain/CrewAI extras.
MCP bridge/server	Public packages and registry listings	Local agent integration and MPS Preflight discovery.
Agent card/offers/context	HTTP 200	Machine-readable capabilities and acquisition boundaries.
llms.txt + sitemap	HTTP 200	Crawler orientation and canonical page discovery.
Bazaar	Context Compiler listed	Autonomous x402 discovery; currently stale relative to live declaration.
Playground and recipes	Public and executable	Zero-install evaluation and copyable integration paths.
Search cluster	Context and MCP guides published	Human search intent and internal linking.

Discovery improvements with the highest expected return

- Add automated contract freshness checks across OpenAPI, SDK types, Bazaar declaration, agent offers, MCP contract, and public examples.
- Make every package and registry listing deep-link to the exact relevant integration page, not only the homepage.
- Measure package-page referral, docs-to-key conversion, first successful API call, second-week repeat usage, and paid conversion.
- Publish customer-shaped examples: long RAG trace, governed MCP risk tool, and one domain-specific GPU workload.
- Avoid more generic SEO pages until Search Console shows an intent with impressions and a product action to satisfy it.

PRIORITIZED BY COMMERCIAL CONSEQUENCE

Material gaps and technical debt

Priori ty	Gap	Consequence	Resolution
P0	Bazaar metadata differs from live x402 declaration	External agents can route using stale schema/description.	Fresh bounded settlement/index refresh; CI drift gate.
P0	No verified recurring external customer usage in repository evidence	Engineering may optimize the wrong workflow.	3-5 labeled pilots; one paid conversion.
P1	MCP gateway routes/registries are not visibly canonical	Security review and support ambiguity.	Consolidation ADR, migration, compatibility window.
P1	QUBO public contract conflicts with promotion documentation	Technical credibility and buyer risk.	Current benchmark/evidence or remove route from public surface.
P1	Protected readiness health not observable in this report	Public 401 is correct but does not prove dependencies green.	Attach current signed/authorized health artifact to releases.
P1	Credit abstraction spans unrelated work units	Buyers cannot predict value or cost.	Usage calculator and product-specific examples.
P2	Single deployment contains public site, payments, operator APIs, and workloads	Large release blast radius.	Split only after load/team evidence justifies it.
P2	No enterprise trust packet	Procurement friction despite strong controls.	Security, privacy, subprocessors, SLA, incident, DPA materials.
P3	Standalone /.well-known/x402 absent	Potential missed non-Bazaar discovery convention.	Add only if ecosystem clients use it; not required for current flow.

Do not mistake completeness for progress

The platform already has more capability than its validated demand. The next increment should reduce buyer friction, create proof, or fix a credibility defect. Everything else belongs in a demand-gated backlog.

RANKED BY PROBABILITY AND STRATEGIC VALUE

Revenue paths

Ran k	Path	Buyer	Offer	ARPA	Near-term probability
1	Context Compiler pilot -> API plan	Developers with costly RAG/tool context	Low-friction benchmark; measurable savings	Medium	High
2	Governed MCP pilot	Security/platform teams deploying MCP	Implementation + control evidence	High	Medium
3	MPS document workflow	Publishers, research, policy, communications	Human-authorized high-consequence review	Medium	Medium
4	Specialist GPU API	Optimization/robotics teams with exact fit	Fast bounded solver without operating Modal	Medium	Low
5	Organic x402 microcalls	Autonomous agents	Zero-contract machine purchase	Low per call	Uncertain
6	SDK/registry-led inbound	Framework users	Convenient integration	Indirect	Medium

Recommended packaging

Offer	Scope	Success evidence	Suggested commercial structure
Context efficiency pilot	10-20 sanitized labeled traces; one downstream model/workflow.	Retention, task quality, p95, total cost per successful task.	Free or small fixed fee credited to first subscription.
Governed MCP pilot	One tenant, 2-5 servers, named tools, 30 days.	Blocked disallowed call, audit review, failure containment, operator feedback.	Paid implementation + monthly platform fee; quote after discovery.
MPS review workflow	One document type and editorial owner.	Review time, unsupported claims found, repeatability, accepted output.	Per-document or prepaid pack; human authorization.
GPU feasibility sprint	One bounded optimization/registration workload.	Correct objective/residual, E2E latency, infrastructure cost.	Paid technical evaluation, then usage contract.

WHY X402 ALONE IS NOT THE WHOLE MODEL

Revenue economics

Monthly x402 calls	Gross revenue at \$0.001	Interpretation
1,000	\$1	Useful technical validation; commercially immaterial.
10,000	\$10	Evidence of repeated machine discovery, not a business.
100,000	\$100	Meaningful product signal; still small revenue.
1,000,000	\$1,000	Requires sustained high-volume integration or aggregation.
10,000,000	\$10,000	A real microtransaction business with material reliability/cost requirements.

MCRB-1's mean avoided input cost is \$0.014815 at a \$3/M-token reference price. The \$0.001 fee is about 6.75% of that gross avoided input cost and leaves \$0.013815 net input savings before any downstream output cost. This is favorable unit value on the benchmark, but it does not automatically create distribution.

Subscription comparison

Target MRR	Builder at \$20	Scale at \$100	Equivalent x402 calls
\$1,000	50 tenants	10 tenants	1,000,000 calls
\$5,000	250 tenants	50 tenants	5,000,000 calls
\$10,000	500 tenants	100 tenants	10,000,000 calls

Strategic conclusion

Use x402 as a frictionless usage rail and ecosystem differentiator, not the sole go-to-market strategy. Convert repeated users into API plans, and use successful MCP pilots to create higher-value recurring revenue.

VALUE, EFFORT, AND EVIDENCE GATE

Improvement priority matrix

Improvement	Value	Effort	When	Evidence
Refresh Bazaar record + automated drift check	Very high	Low	Immediate	Live warning exists
Customer-context pilot harness and report template	Very high	Medium	Immediate	Existing measurement code
Canonical MCP gateway ADR and migration	High	Medium	0-30 days	Current dual surfaces
QUBO status/evidence reconciliation	High	Low-Medium	0-30 days	Public/doc contradiction
Enterprise trust packet	High	Medium	30-60 days	Needed for procurement
Acquisition-to-retention dashboard	High	Medium	0-45 days	Meters already exist
Customer-specific answer-quality benchmark	High	Medium	Per pilot	Requires labels
MCP private connectivity	Potentially high	High	Demand-gated	Only with design partner
MCP SSE streaming	Unknown	Medium-High	Demand-gated	No verified buyer need
More GPU solvers	Unknown	High	Defer	No validated demand
More generic SEO pages	Low	Medium	Defer	Search intent must lead to action

Decision rule

Ship immediately when the work fixes a live transaction/discovery defect or enables a scheduled pilot. Demand-gate anything that creates a new protocol surface, runtime, or support obligation.

FROM PLATFORM READINESS TO FIRST REPEAT REVENUE

30-60-90 day operating plan

Days 0-30: repair discovery and focus the offer

- Refresh Bazaar metadata and store matching live/catalog digests as release evidence.
- Resolve the QUBO public/private evidence contradiction.
- Publish one canonical architecture/data-flow page and one MCP gateway contract path.
- Recruit 3-5 Context Compiler pilots from teams with long RAG/tool payloads; agree labels and thresholds before testing.
- Define one paid governed-MCP pilot and send it only to buyers with active MCP deployment risk.
- Instrument first discovery, key creation, first successful call, repeated call, checkout, and renewal.

Days 31-60: convert proof into a repeatable sale

- Complete pilot reports with required-fact retention, downstream task result, E2E p95, and cost per successful task.
- Convert at least one successful Context pilot to Builder/Scale or a negotiated usage plan.
- Start one paid MCP pilot; document allowlist policy, incident simulation, and weekly audit review.
- Publish an anonymized case only with customer permission and stated limitations.
- Create a security/procurement packet and a simple data-processing summary.

Days 61-90: expand only from demonstrated pull

- If Context usage repeats, add workload templates and volume pricing; if not, revise the target job rather than the algorithm.
- If MCP pilot usage is active, scope the highest requested enterprise control: OAuth, private connectivity, or streaming - one only.
- If a GPU design partner exists, publish its workload-specific benchmark and price from actual cost/margin.
- Stop or archive offers with no qualified lead, successful pilot, or repeat use.

METRICS THAT SHOULD DRIVE THE NEXT BUILD

Measurement system

Funnel layer	Core metrics	Cadence
Discovery	Impressions by capability; registry views; docs landing; agent-card/Bazaar fetches	Weekly
Activation	Key created; first successful call; time-to-first-call; playground completion	Weekly
Value	Token cost avoided; required-fact retention; downstream success; operator time saved	Per pilot/workflow
Retention	Second-day, second-week, and fourth-week active tenant; calls per active tenant	Weekly/monthly
Revenue	Gross/net by rail and offer; paid tenants; expansion; refund/dispute; margin	Weekly/monthly
Reliability	Availability, p95/p99, cold start, queue delay, upstream errors, payment failures	Daily/release
Security	Blocked tools/methods, auth failures, circuit opens, replay attempts, secret rotations	Weekly/incident
Discovery freshness	Live vs catalog digest, crawler status, schema validation, last refresh age	Every contract change

North-star recommendation

Paid successful workflows per week

Count a workflow only when a customer or external agent pays, the task completes inside its accepted latency/error boundary, and the customer repeats or explicitly accepts the result. Raw calls, credits consumed, impressions, and compression percentage are supporting metrics - not the business outcome.

Minimum weekly review

- What source produced each qualified user?
- What job were they trying to complete?
- Where did activation fail?
- Which users repeated without operator help?
- Which failures cost money or trust?
- What single build would improve the highest-value stage next week?

SMALL TESTS BEFORE MAJOR BUILDS

Revenue experiments

Experiment	Sample	Success	Stop condition
Context pilot conversion	5 qualified teams; 10-20 traces each	At least 2 complete; 1 converts paid	Stop after 5 if no measurable value
Playground activation	100 qualified visitors	>=10 sample runs; >=3 copy code/key starts	Fix friction before buying traffic
x402 external use	Fresh catalog + 30-day observation	>=1 external payer and >=10 repeat paid calls	Treat absence as distribution signal
MCP design partner	10 targeted security/platform conversations	2 discovery calls; 1 paid pilot	Do not add OAuth/SSE without commitment
MPS document workflow	3 document owners	1 paid repeat workflow or clear rejection reason	Do not broaden generic claims
GPU workload fit	5 domain-specific prospects	1 bounded dataset and paid feasibility sprint	No new solver without data

Commercial discipline

A failed experiment is useful when it names the buyer, offer, sample, channel, threshold, and stop condition. It prevents the platform from responding to weak demand by adding unrelated features.

PROTECT FOCUS AND OPERATING CAPACITY

What not to build yet

- Another general AI endpoint without a named buyer, labeled workload, and payment model.
- More GPU engines because they are technically interesting; require a design partner and benchmark corpus.
- A second dashboard or billing abstraction; improve the existing activation and cost explanation first.
- Cross-chain x402 support before Base usage is material and compliance/accounting is settled.
- Custodial wallets, automated asset conversion, or agent spending policy without specialist legal and security design.
- SSE in the MCP Gateway without a customer requiring long-lived streaming and agreeing on timeout/billing semantics.
- Private networking as a generic platform feature before an enterprise prospect identifies the required topology.
- Seven more thin search pages; improve existing pages from Search Console demand and real benchmark evidence.
- Customer-facing SLA promises that exceed one operator's support and escalation capacity.

The correct instinct

Suppressing speculative building is rational. The platform is sufficiently complete to sell. The next build should be pulled by a live defect, a measurable activation bottleneck, or a buyer commitment.

QUESTIONS FOR THE NEXT STRATEGY SESSION

Brainstorming decision agenda

Decision	Question
Product	Which single job will Maha optimize for the next 30 days?
Buyer	Who already feels that problem weekly and controls a budget?
Proof	What labeled input and accepted output will prove value?
Offer	What can be bought without interpreting the entire platform?
Rail	Should the buyer use Stripe/API credits, x402, or a paid pilot - and why?
MCP	Is the gateway primarily a SaaS product, an implementation service, or both?
Pricing	Does the price track value, compute cost, risk reduction, or operator effort?
Support	What availability and response promise can one operator sustain?
Compliance	What does legal counsel require for x402 accounting, disclosure, and geographic scope?
Stop	Which capabilities will be archived if they attract no qualified demand in 90 days?

Recommended starting decision

Select the Context Compiler customer-context pilot as the activation wedge, and the governed MCP pilot as the enterprise expansion offer. Everything else supports those two paths or remains demand-gated.

OPENAPI 0.1 - 35 PATHS

Appendix A - Public API inventory

Domain	Paths
Identity/billing	POST keys/generate, keys/checkout, keys/rotate, keys/revoke; GET keys/balance; subscription/settings
Context	POST /api/v1/compress; POST /api/context-packs; POST /api/context-pack-evaluations
GPU jobs	POST qubo-ising, tensor-network, geometric-registration; GET job by ID
MCP	Register, list, discover, set policy, settings, gateway proxy
Audit	MPS audits, credit balance/checkout/webhook, public audit, audit export
Claims/chat	Claim lookup and OpenAI-compatible chat surface
Books	Entitlement, content, authenticated/public checkout, webhook
Discovery	Agentic offers and MCP bridge manifest

The platform also contains private admin/operator routes for revenue, billing readiness, observability, market mapping, search performance, experiments, operations, content workflow, and MCP administration. These should remain outside the public OpenAPI unless a customer contract requires them.

Contract governance recommendation

- Retitle the public contract from 'MPS API' to the broader Maha Developer Platform while preserving the API version and existing paths.
- Assign each path an owner, maturity label, authentication mode, payment mode, data-retention boundary, and deprecation policy.
- Generate SDK conformance tests from OpenAPI or assert schema parity in CI to prevent hand-authored drift.

PRIMARY REPOSITORY AND LIVE SOURCES

Appendix B - Evidence register

Evidence area	Source
Repository	Commit 32976b3 on main; clean worktree at report start.
Architecture	README.md; .env.example; app/api route inventory; lib/* control modules.
OpenAPI	lib/openapi.ts and live /api/docs/openapi (HTTP 200).
Context benchmark	benchmarks/mcrl-1/results.json, cases.jsonl, cohort.json; benchmark page.
GPU evidence	docs/benchmarks/restored-engines-a10g-baseline-2026-08-06.md; qubo baseline and promotion docs.
Payments	lib/x402/config.ts, gateway.ts, protocol.ts, replay-guard.ts, chain.ts; Stripe billing modules.
MCP	lib/mcp/*, lib/mcp-gateway.ts, app/api/v1/mcp/*, app/api/mcp-gateway/*, public contract.
Operations	docs/observability.md, release-recovery.md, slo-capacity.md, database-migrations.md; 11 workflows.
Migration proof	docs/production-migration-evidence-2026-08-03.md; 66 migration files.
Developer ecosystem	packages/sdk 0.2.0; clients/python 0.1.0; public discovery documents; recipes/playground.
Live checks	Public pages/discovery HTTP checks and x402 Doctor read-only run at 2026-08-08T09:00Z.
Validation	420/420 tests, TypeScript, ESLint, and Next.js production build passed in the immediately preceding validated repository state.

Important evidence limitations

- This report did not use protected operator tokens, so it does not independently certify current billing, revenue, or observability readiness as green.
- No new paid settlement was executed. Historical production settlements were supplied by the operator during the working session.
- Public package and registry publication states were inferred from repository links and package metadata; third-party listing analytics were not available.
- Revenue scenarios are arithmetic illustrations, not forecasts.
- Security assessment is code- and design-based, not a penetration test or formal audit.

ONE SENTENCE

Recommended next move

Focus the next 30 days

Repair Bazaar freshness, resolve the QUBO evidence contradiction, then convert labeled Context Compiler pilots and one governed MCP design partner - do not expand the product catalog.

This sequence protects technical credibility, improves autonomous discovery, creates customer-specific proof, and preserves the MCP Gateway as the higher-value expansion path. It also gives legal counsel a bounded x402 operating model to review rather than an open-ended crypto strategy.

End of report